

FPGA Internship Evaluation – 12-Hour Lab Assessment

Overview

This 12-hour lab assessment is designed to evaluate candidates for an FPGA internship through hands-on RTL design, simulation, and synthesis workflows using open-source EDA tools. Participants will work independently using their own cloud environment and demonstrate both technical understanding and documentation skills.

Test Format

- **Duration:** 12 hours (continuous)
- **Mode:** Self-paced, hands-on lab execution
- **Resources Provided:**
 - Access to lab video content
 - Links to cloud-based lab environments

Participant Requirements

Each participant must:

1. Use their **own cloud account/environment** for all lab activities.
2. Execute all labs independently using provided video guidance.
3. Capture **waveform outputs (GTKWave screenshots)** and synthesis outputs where applicable.
4. Upload all work to a **personal GitHub repository**, including:
 - Waveform snapshots
 - RTL code and synthesis scripts
 - Clear explanations of each lab and observations

Evaluation Criteria

Participants will be assessed based on:

- Correctness of simulations and synthesis results
- Understanding of RTL design and digital logic concepts
- Quality of waveform analysis and debugging
- Clarity and depth of explanations
- Organization and completeness of GitHub submission

Lab Topics & Modules

1. Day 1 – RTL Simulation & Waveform Analysis

- Introduction to RTL design and simulation flow
- Writing Verilog modules and testbenches
- Compiling and simulating using Icarus Verilog (iverilog)
- Generating and analyzing waveforms using GTKWave
- Running basic labs involving simulation and debugging

2. Day 2 – Logic Synthesis & Optimization

- Introduction to synthesis using Yosys
- Understanding synthesis flow and generated netlists
- Working with SKY130 standard cell libraries
- Performing synthesis for simple RTL designs (e.g., mux)
- Observing synthesis results and optimizations
- Introduction to timing libraries (.lib) and basic concepts

Submission Guidelines

- Submit a **GitHub repository link** within the allotted time.
- Repository must be well-structured with clear folder organization.
- Include a **README.md** summarizing:
 - Labs completed
 - Key learnings
 - Observations and challenges

Note: This assessment is intended to evaluate practical FPGA workflow readiness, including RTL design, simulation, synthesis, debugging, and technical documentation skills.